

Response to the reviewer

Revised manuscript: *Capacity- and Compute-Matched Evaluation of Published BU-Net Components with Multi-Seed Development and Independent External Testing for Multimodal Glioma Segmentation*

We agree that the rejected manuscript could not be repaired through language editing or a few additional experiments. The study was rebuilt as a controlled evaluation of published components. This response is a result-bound template: it may be rendered only after the development matrix, statistical freeze, and single external testing session are complete. Page and line references remain layout placeholders until the final proof is generated.

R01 — BU-Net attribution and novelty

Concern. RES and WC were not attributed to BU-Net and appeared to be positioned as new contributions.

Direct response. We agree. RES and WC are now attributed to Rehman et al. throughout. The revised work does not claim a new architecture.

Exact action. We replaced the novelty framing with a capacity-controlled evaluation question, cited BU-Net as the primary source for both components, and separated conventional residual blocks from RES pathways.

New evidence. The model matrix contains plain, RES, WC, BU-Net, conventional residual-block, parameter-matched, and compute-matched controls.

Manuscript location. [FINAL PAGE/LINES: title, Introduction, Models, and Discussion]; [FINAL TABLE/FIGURE: architecture and model-configuration table].

Repository evidence. literature/q1q2_verified_sources.yaml;

configs/q1q2_v2/model_matrix.yaml; reports/q1q2_v2/model_matrix_validation.json.

Remaining limitation. The contribution is controlled evidence about published components, not architectural invention.

R02 — BraTS edition overlap and canonical cohort

Concern. Combining BraTS editions without patient deduplication could duplicate patients and create leakage.

Direct response. We did not pool BraTS editions. BraTS 2020 is the sole development cohort; BraTS 2019 is used only for identity and content auditing.

Exact action. We created read-only inventories, file/content hashes, cross-edition mappings, and a canonical patient manifest.

New evidence. The revised development cohort contains

[PENDING:DESIGN.DEVELOPMENT_PATIENTS] unique labeled patients. BraTS-Africa contributes [PENDING:DESIGN.EXTERNAL_CONFIRMATORY_PATIENTS] confirmatory glioma patients and [PENDING:DESIGN.EXTERNAL_SUPPORTIVE_PATIENTS] separately reported other-neoplasm patients.

Manuscript location. [FINAL PAGE/LINES: Datasets and Overlap Audit]; [FINAL FIGURE: cohort identity and overlap flow].

Repository evidence. manifests/canonical/brats2020_canonical_manifest.csv;

manifests/audit/duplicate_mapping.csv; reports/q1q2_v2/external_gate_c_summary.json.

Remaining limitation. Content-based auditing reduces identifiable overlap risk but cannot prove that every historical data lineage is documented.

R03 — Patient-level partitions and legacy-test isolation

Concern. The original split was unspecified and might have been made at slice level; the meaning of “test set” was unclear.

Direct response. Every revised partition is patient-level. The previously opened internal subset is retained only as legacy evidence and is unavailable to all new model selection and inference paths.

Exact action. We replaced the old split for v2 development with deterministic patient-level cross-validation and an independent external test. External and legacy accesses are separately guarded and logged.

New evidence. Each development patient appears in exactly one of [PENDING:DESIGN.FOLDS] validation folds, and every main model uses the same [PENDING:DESIGN.SEEDS] training seeds.

Manuscript location. [FINAL PAGE/LINES: Study Design and Leakage Controls].

Repository evidence. splits/q1q2_v2/split_metadata.json;
configs/q1q2_v2/gate_g_freeze.yaml; artifacts/test_access_log.jsonl.

Remaining limitation. The folds support development estimation; they are not relabeled as an official BraTS test.

R04 — Common and fair experimental protocol

Concern. Models were previously compared across different data subsets, losses, schedules, and training opportunities.

Direct response. The heterogeneous historical comparison was removed. All revised comparisons use the same cohort contracts, folds, seed list, evaluator, preprocessing boundaries, and prespecified budget definitions.

Exact action. We froze a shared native protocol, retained official framework behavior where required for nnU-Net, and record every deliberate model-specific exception.

New evidence. All planned runs and failures are retained in the run registry; literature-derived scores are excluded from experimental tables.

Manuscript location. [FINAL PAGE/LINES: Training and Compute Protocol]; [FINAL TABLE: training and tuning budgets].

Repository evidence. configs/q1q2_v2/training_protocol.yaml;
configs/q1q2_v2/model_matrix.yaml; configs/q1q2_v2/gate_g_freeze.yaml.

Remaining limitation. Fair comparison does not mean that official self-configuring systems and native implementations have identical optimization internals.

R05 — Convergence and prior undertraining

Concern. Short or unequal training, especially the prior Swin experiment, could not support conclusions about architecture.

Direct response. The prior short-run curves are legacy evidence only. They do not appear in revised architecture comparisons.

Exact action. We prespecified convergence checks, validation frequency, early-stopping behavior, maximum steps, terminal and best checkpoints, and separate compute-matched runs. Failed convergence remains visible.

New evidence. The final convergence table and ranking-stability figure are generated only from complete run artifacts; an incomplete model cannot be interpreted as an architectural failure.

Manuscript location. [FINAL PAGE/LINES: Training and Compute Protocol]; [FINAL FIGURE: convergence and ranking stability].

Repository evidence. configs/q1q2_v2/training_protocol.yaml;
configs/protocols/convergence_matched_mps.yaml; configs/q1q2_v2/gate_g_freeze.yaml.

Remaining limitation. Convergence is operationally defined under the frozen device and optimization protocol, not proven in an asymptotic sense.

R06 — Strong experimental baselines

Concern. Direct BU-Net, residual U-Net, nnU-Net, volumetric, and modern hybrid baselines were missing.

Direct response. The revised matrix contains the component core, official nnU-Net configurations, a controlled five-slice model, and a maintained Swin UNETR implementation.

Exact action. We validate all models against one data/evaluator contract and prohibit replacing an unexecuted comparator with a literature score.

New evidence. The frozen comparison matrix contains [PENDING:DESIGN.MODELS] models and [PENDING:DESIGN.DEVELOPMENT_RUNS] development runs.

Manuscript location. [FINAL PAGE/LINES: Models and Component Attribution]; [FINAL TABLE: model configurations].

Repository evidence. configs/q1q2_v2/model_matrix.yaml;
reports/q1q2_v2/model_matrix_validation.json;
reports/q1q2_v2/nnunet_planning_summary.json.

Remaining limitation. The chosen modern baselines do not exhaust the rapidly changing segmentation literature.

R07 — Component ablation and capacity attribution

Concern. RES, WC, residual blocks, and model capacity were not isolated.

Direct response. We agree that the historical comparison was capacity-confounded. The revised primary estimand compares U-Net+RES with a plain parameter-matched U-Net.

Exact action. We added same-width, parameter-matched, compute-matched, WC, BU-Net, conventional residual-block, and interaction controls.

New evidence. For the primary external contrast, U-Net+RES achieved [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.FIRST_MEAN] and the matched plain U-Net achieved [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.SECOND_MEAN]. The paired difference was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.MEAN_DIFFERENCE] with interval [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_BOOTSTRAP_LOWER_95] to [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_BOOTSTRAP_UPPER_95]. [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.INTERPRETATION_TEXT]

Manuscript location. [FINAL PAGE/LINES: Primary Capacity-Controlled Comparison]; [FINAL TABLE: primary contrasts].

Repository evidence. reports/q1q2_v2/model_matching_summary.json;
configs/q1q2_v2/statistical_analysis_plan.yaml;
artifacts/q1q2_v2/statistics/primary_contrasts.csv.

Remaining limitation. The primary conclusion applies to the implemented RES definition and frozen capacity target, not every residual architecture.

R08 — Loss definition, output semantics, and loss ablation

Concern. The loss equation, activations, class inclusion, and selection procedure were incomplete.

Direct response. The revised loss catalog defines every activation, channel, reduction, class inclusion rule, parameter, smoothing term, and empty-class behavior.

Exact action. Development-only comparison covers softmax-consistent and hybrid candidates. One loss is frozen before main runs, and a prespecified architecture-by-loss sensitivity is kept separate from the primary estimand.

New evidence. The selected objective is **[PENDING:METHOD.SELECTED_LOSS]**; external labels and the legacy internal subset are prohibited from loss selection.

Manuscript location. [FINAL PAGE/LINES: Loss Selection and Optimization]; [FINAL SUPPLEMENT: executable loss definitions and interaction analysis].

Repository evidence. configs/losses/catalog.yaml; configs/q1q2_v2/loss_protocol.yaml;
reports/q1q2_v2/loss_methods.json.

Remaining limitation. The loss screen is intentionally restricted and does not establish that the selected loss is universally optimal.

R09 — Complete central evaluation and empty-mask behavior

Concern. Dice/IoU reporting was inconsistent, empty slices were unclear, and boundary and lesion endpoints were missing.

Direct response. All revised predictions pass through one evaluator. Validation and external testing retain complete volumes, including empty slices.

Exact action. We froze regional overlap, boundary, surface, rate, volume, lesion-detection, lesion-wise, and false-positive endpoints with explicit empty-mask and infinite-distance rules.

New evidence. The final tables report WT, TC, and ET separately and retain one-empty HD95 as infinity with its rate shown alongside finite summaries.

Manuscript location. [FINAL PAGE/LINES: Central Evaluation]; [FINAL TABLES: regional and lesion-wise endpoints].

Repository evidence. configs/q1q2_v2/evaluation.yaml;
configs/q1q2_v2/evaluation_sensitivity.yaml; src/evaluation.

Remaining limitation. Reference annotations remain imperfect proxies for the underlying biological boundaries.

R10 — Equal seeds and patient-level statistical inference

Concern. Replication counts, uncertainty, paired tests, effect sizes, and multiplicity control were absent.

Direct response. Every main model now shares the same folds and seeds. The patient is the inferential unit.

Exact action. We prespecified paired patient bootstrap intervals, hierarchical seed-patient resampling, paired sign-flip tests, Holm correction, effect size, and probability-of-superiority reporting.

New evidence. The primary contrast used

[PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIREDPATIENT_COUNT] paired external patients. Its Holm-adjusted p value was

[PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.HOLM_ADJUSTED_P] and probability of superiority was

[PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PROBABILITY_OF_SUPERIORITY].

Manuscript location. [FINAL PAGE/LINES: Statistical Analysis and Primary Results]; [FINAL FIGURE: hierarchical uncertainty].

Repository evidence. configs/q1q2_v2/seeds.yaml;

configs/q1q2_v2/statistical_analysis_plan.yaml; artifacts/q1q2_v2/statistics/completion.json.

Remaining limitation. Statistical precision is bounded by the available independent external patients and does not imply clinical importance.

R11 — Measured resource and efficiency evidence

Concern. Resource-efficiency claims lacked parameters, computation, memory, and latency measurements.

Direct response. Unqualified efficiency and deployment language was removed. Costs are reported as separate measured dimensions.

Exact action. We profile parameters, graph computation, accelerator time, preprocessing, forward, post-processing, end-to-end latency, and backend-appropriate memory counters under a frozen protocol.

New evidence. U-Net+RES contains

[PENDING:RESOURCE.UNET_RES.PARAMETER_COUNT] parameters, requires

[PENDING:RESOURCE.UNET_RES.FLOPS_PER_DECLARED_INPUT] FLOPs per declared input, and has median external end-to-end latency

[PENDING:RESOURCE.UNET_RES.INFERENCE_END_TO_END_P50_SECONDS] seconds per volume. Peak framework-allocated unified memory is

[PENDING:RESOURCE.UNET_RES.TRAINING_PEAK_FRAMEWORK_ALLOCATED_UNIFIED_MEMORY_BYTES_MAX] bytes.

Manuscript location. [FINAL PAGE/LINES: Resource Analysis and Resource Realization];

[FINAL FIGURES: capacity and compute Pareto views].

Repository evidence. configs/q1q2_v2/resource_profile_protocol.yaml;

configs/q1q2_v2/resource_execution.yaml;

artifacts/q1q2_v2/resources/accuracy_cost_pareto.csv.

Remaining limitation. Apple MPS measurements describe one unified-memory system and are not CUDA throughput or VRAM measurements.

R12 — Controlled dimensionality and transformer context

Concern. The original manuscript compared two-dimensional and volumetric or transformer systems without a common protocol.

Direct response. Heterogeneous literature scores and subjective rankings were removed from experimental figures.

Exact action. Native two-dimensional, controlled five-slice, official nnU-Net volumetric, and maintained Swin UNETR systems are evaluated on common patients through the central evaluator.

New evidence. Dimensionality and architecture results are reported as model-specific estimates and paired comparisons where prespecified, not as a claim that one family is universally preferable.

Manuscript location. [FINAL PAGE/LINES: Models, Results, and Discussion].

Repository evidence. configs/q1q2_v2/model_matrix.yaml;
reports/q1q2_v2/nnunet_planning_summary.json;
reports/q1q2_v2/m1_execution_decision.json.

Remaining limitation. Dimensionality, implementation ecosystem, and optimization cannot be separated by a single contrast.

R13 — Independent external testing and domain shift

Concern. The original work lacked an independent hospital/domain test and could not support generalization or clinical claims.

Direct response. The legacy internal subset is not reused. BraTS-Africa provides the independent domain-shift cohort after a zero-overlap audit.

Exact action. We permit one audited external session only after model, checkpoint, endpoint, multiplicity, and analysis freeze. No external tuning, retraining, or label-informed adaptation is allowed.

New evidence. The external session evaluates

[PENDING:DESIGN.EXTERNAL_MAIN_CHECKPOINTS] frozen checkpoints. The confirmatory

cohort contains [PENDING:DESIGN.EXTERNAL_CONFIRMATORY_PATIENTS] glioma patients; other neoplasms are reported separately as supportive evidence.

Manuscript location. [FINAL PAGE/LINES: External Evaluation and External Results]; [FINAL FIGURE: external subgroup analysis].

Repository evidence. reports/q1q2_v2/external_gate_c_summary.json;
configs/q1q2_v2/gate_h_external.yaml; artifacts/q1q2_v2/external_access_log.jsonl.

Remaining limitation. External testing on one public African cohort does not establish universal transportability or clinical utility.

R14 — Qualitative success and failure analysis

Concern. The original work showed no segmentation examples, overlays, or hard failures.

Direct response. The revised design includes success, median, lesion, false-positive, boundary, and model-disagreement cases.

Exact action. Case identities are selected after evaluation by frozen deterministic rules. Panels include all modalities, reference labels, all models, lesion components, and false-positive/false-negative overlays.

New evidence. Both favorable and adverse cases remain in the final panel set; identities are not manually substituted.

Manuscript location. [FINAL PAGE/LINES: Qualitative Analysis]; [FINAL FIGURE: qualitative panels].

Repository evidence. configs/q1q2_v2/qualitative_protocol.yaml;
configs/q1q2_v2/qualitative_execution.yaml; artifacts/q1q2_v2/qualitative/selected_cases.json.

Remaining limitation. Selected examples illustrate measured failure modes; they do not estimate their population prevalence.

R15 — Artifact-derived figures and tables

Concern. Heterogeneous scores and subjective cost or architecture ratings were presented as quantitative comparisons.

Direct response. Those figures and tables were discarded.

Exact action. Every revised quantitative panel is generated from the central metrics, statistical, subgroup, resource, or qualitative artifacts. No literature score is plotted beside experimental output.

New evidence. Figure manifests retain input hashes, output hashes, and generation metadata; table values are generated rather than manually copied.

Manuscript location. [FINAL PAGE/LINES: Results and figure captions]; [FINAL SUPPLEMENT: output manifest].

Repository evidence. configs/q1q2_v2/figure_execution.yaml;
src/bratsarticle/analysis/q1q2_figures.py; artifacts/q1q2_v2/figures/completion.json.

Remaining limitation. Visual presentation choices remain editorial even when the plotted values and selected cases are artifact-bound.

R16 — Primary literature coverage

Concern. Essential primary sources and critical comparisons were missing.

Direct response. The literature ledger now includes BU-Net, Focal Tversky, nnU-Net and its controlled reevaluation, TransBTS, nnFormer, Swin UNETR, dimensionality studies, Metrics Reloaded, CLAIM, seed variability, and the BraTS-Africa descriptor.

Exact action. Bibliographic metadata and contribution fields were checked against primary publisher, proceedings, author-manuscript, or official data pages. Unverified fields remain explicitly marked.

New evidence. The literature matrix separates datasets, dimensionality, seed reporting, overlap auditing, external evaluation, metrics, code, and license instead of treating published scores as a benchmark.

Manuscript location. [FINAL PAGE/LINES: Introduction, Related Work, and References];
[FINAL SUPPLEMENT: novelty matrix].

Repository evidence. literature/q1q2_verified_sources.yaml;
reports/q1q2_v2/novelty_matrix.csv; reports/q1q2_v2/novelty_assessment.md.

Remaining limitation. The manuscript is an experimental methods study, not a systematic review, and does not claim exhaustive literature coverage.

R17 — Reproducibility, code, environment, and license

Concern. Code, splits, configurations, environment details, and a reproducibility package were absent.

Direct response. The repository now contains versioned code, configurations, manifests, tests, environment locks, audit contracts, and an Apache license.

Exact action. Gate I verifies source cleanliness, tests, static analysis, hashes, downstream numerical regeneration, standard figures, and qualitative panel hashes. Full retraining and artifact-only reproduction are described as different scopes.

New evidence. Final reproduction status is reported from the Gate I completion artifact; a clean-clone audit is not represented as full retraining.

Manuscript location. [FINAL PAGE/LINES: Reproducibility and Code/Data Availability]; [FINAL TABLE: reproducibility checklist].

Repository evidence. REPRODUCIBILITY.md;
configs/q1q2_v2/reproducibility_execution.yaml; LICENSE.

Remaining limitation. Authorized raw data remain subject to source access terms and are not redistributed.

R18 — Writing, organization, and bounded claims

Concern. The original text was repetitive, unnatural, and made causal, clinical, or architecture-wide claims beyond the evidence.

Direct response. The manuscript was rewritten as one experimental study with a structured abstract, explicit estimand, compact Methods, result-led Discussion, and bounded Conclusion.

Exact action. We removed tutorial taxonomy, pseudo-quantification, literature-score rankings, unsupported mechanism claims, and clinical applicability language. Gate J audits result provenance and inferential wording.

New evidence. The final primary interpretation is:

[PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.INTERPRETATION_TEXT]

Manuscript location. [FINAL PAGE/LINES: complete revised manuscript].

Repository evidence. manuscript/q1q2_v2_manuscript.template.md;
configs/q1q2_v2/claim_execution.yaml; claims/q1q2_v2_claim_ledger.csv.

Remaining limitation. Professional language editing cannot substitute for author review of scientific meaning, and all authors must approve the final wording.

R19 — Submission declarations and exact response parity

Concern. The revised response must not claim results or resource fields that are absent from the manuscript and submission package.

Direct response. The response and manuscript use the same claim registry. Every rendered value records its source file, hash, selector, column, format, and rendered-document hash.

Exact action. Final page/line references are added only after layout. Author order, affiliations, contributions, ethics, funding, conflicts, acknowledgements, correspondence details, and disclosure text require author confirmation.

New evidence. Gate J must pass for both the manuscript and this response; Gate K remains blocked until the author declarations and journal-specific files are complete.

Manuscript location. [FINAL PAGE/LINES: declarations and availability sections].

Repository evidence. configs/q1q2_v2/claim_execution.yaml;

submission/AUTHOR_CONFIRMATION_REQUIRED.md;

artifacts/q1q2_v2/claims/completion.json.

Remaining limitation. No author declaration, journal choice, release DOI, or page/line citation is inferred from repository code.